Worksheet 2-5 & 2-6 B
Tangent Lines, Normal Lines and Piecewise Functions #1
AP Calculus Name_____

Find the equation of the tangent line and the equation of the normal line to the function at the given point.

1.) $f(x) = \sqrt{x}$ at $f(9) = 3$

2.) $g(x) = x^3 - x$ at $x = 1$

3.) If $f(2) = 3$ and $f'(2) = 5$, find the equations of the tangent line and the normal line to the graph of $f(x)$ at the point where $x = 2$

4.) Is the function differentiable at the given value of x?

$$f(x) = \begin{cases} \frac{30}{\sqrt{x}} - x^2 & x < 9 \\ x^2 - 5x - 107 & x \geq 9 \end{cases} \text{ at } x = 9$$

5.) What values of a and b would make the function differentiable at $x = -1$?

$$f(x) = \begin{cases} x^3 - ax + 4 & x \leq -1 \\ ax^2 + bx - 2 & x > -1 \end{cases}$$